A10.1 Interoperability principles

The following interoperability principles apply to both AXI3 and AXI4 components.

As a general principle, components must support all combinations of inputs, but do not have to generate all combinations of outputs. For example, a slave must support all the different possible lengths of burst, but a master only has to generate the types of burst that it uses. This policy ensures that all components work with all other components.

The conditions under which a signal can be omitted from an AXI interface are:

Optional Outputs

If a component might require a value that does not match the default value, then the component must have the output signal present.

If a component always requires the value that matches the default value, specified in *Default signal values* on page A10-112, then it is not required that the component has the signal present.

Optional Inputs

An input signal can be omitted if the master or slave does not need to observe the input signal for correct functional operation.

Note

Interconnect components can also omit signals when appropriate. For example, when a signal is only ever driven to its default value, there is no requirement to transport that signal across the interconnect. The signal can be created at its destination. Similarly, if a signal is not used at any destination then there is no requirement to transport it across the interconnect.

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